MINI-THESIS

Research Methodology

Effectiveness of Climate Policies : Statistical Analysis of the Impact of the Carbon Tax on CO2 Emissions and Economic Growth in Sweden (1981–2020)

Student :
Jichi Sosthène

Professor :
Dr. Kouadio Bertin

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1 Introduction

Climate change is now considered one of the major challenges facing contemporary economies. The continuous increase in greenhouse gas emissions, particularly carbon dioxide (CO2), is widely recognized as the main driver of global warming. In response to this situation, governments have progressively implemented public policies aimed at reducing the carbon footprint of their economies while maintaining economic growth.

Among these policy instruments, the carbon tax occupies a central position. It is based on the principle that economic agents should bear the social cost of the pollution they generate. However, this fiscal mechanism has sparked considerable debate. Some economists and political actors argue that such a tax could reduce business competitiveness, slow down economic growth, and negatively affect household purchasing power.

Sweden represents a particularly relevant case study. Indeed, the country introduced one of the world's first carbon taxes as early as 1991, while simultaneously achieving sustained long-term economic growth and a significant reduction in CO2 emissions. This apparent paradox makes Sweden an ideal laboratory for analyzing the compatibility between ambitious environmental policy and economic performance.

This mini-thesis therefore aims to empirically analyze the impact of the Swedish carbon tax on CO2 emissions and economic growth over the period 1981–2020.

2 Research problem

One of the fundamental debates in environmental economics revolves around the following question :

> *Is it possible to sustainably reduce CO2 emissions without compromising economic growth ?*

In many countries, economic growth has historically been associated with an increase in fossil energy consumption and, consequently, higher carbon emissions. This relationship raises important questions about the actual effectiveness of climate policies based on environmental taxation.

Thus, the central research problem of this study can be formulated as follows : **Has the carbon tax implemented in Sweden succeeded in reducing CO2 emissions while maintaining, or even promoting, economic growth ?**

3 Research objectives

3.1 General objective

The main objective of this mini-thesis is to assess the effectiveness of the Swedish carbon tax by analyzing its impact on CO2 emissions and long-term economic growth.

3.2 Specific objectives

More specifically, this study aims to :

- analyze the evolution of CO2 emissions per capita before and after the introduction of the carbon tax in 1991 ;
- examine the dynamics of GDP per capita over the same period ;
- measure the statistical relationship between economic growth and CO2 emissions ;
- highlight the possible existence of a decoupling phenomenon between economic growth and environmental pollution.

4 Literature review and theoretical framework

4.1 Pigouvian tax

The carbon tax is grounded in the theoretical framework of the Pigouvian tax, developed by economist Arthur Pigou. According to this approach, when an economic activity generates a negative externality, such as pollution, the market alone fails to achieve an optimal allocation of resources. The introduction of a tax allows the social cost of this externality to be internalized.

In the case of carbon emissions, the tax aims to increase the cost of fossil fuels in order to encourage economic agents to reduce polluting consumption and invest in cleaner technologies.

4.2 Environmental Kuznets curve

Another key theoretical framework mobilized in this study is the Environmental Kuznets Curve (EKC). This hypothesis suggests that at low income levels, economic growth is accompanied by increased pollution. However, beyond a certain level of development, the relationship reverses, as societies invest more in environmental protection, leading to a reduction in emissions.

The Swedish experience makes it possible to empirically test this hypothesis by examining whether the carbon tax accelerated the transition toward greener economic growth.

5 Research methodology

5.1 Nature of the study

This study adopts a quantitative approach based on the analysis of secondary data obtained from reliable international sources.

5.2 Data sources

The data used cover the period 1981–2020 and mainly come from :

- the World Bank (GDP per capita and CO2 emissions per capita) ;
- international climate and environmental databases.

5.3 Variables analyzed

- CO2 emissions per capita (in metric tons)
- GDP per capita (in constant US dollars)

5.4 Analytical tools

The statistical analysis relies on :

- descriptive statistics ;
- before-and-after comparisons relative to 1991 ;
- Pearson correlation coefficient ;
- graphical representations such as line charts, scatter plots, and boxplots.

6 Detailed presentation of results

This section presents, in a structured and progressive manner, the empirical results derived from the analysis of data covering the period 1981–2020. The objective is not merely to describe numerical values, but to highlight major trends and significant structural breaks observed in the Swedish economy.

6.1 Evolution of CO2 emissions per capita

The data analysis reveals a contrasted evolution of CO2 emissions per capita in Sweden. In the early 1980s, emissions were relatively high, averaging around 8 metric tons per capita. This situation reflects an economy still largely dependent on fossil fuels, despite an already advanced level of economic development.

From 1991 onwards, the year the carbon tax was introduced, a clearly downward trend can be observed. Emissions declined progressively and continuously, reaching approximately 3.5 metric tons per capita in 2020. This decrease represents a reduction of more than 50% over the post-tax period.

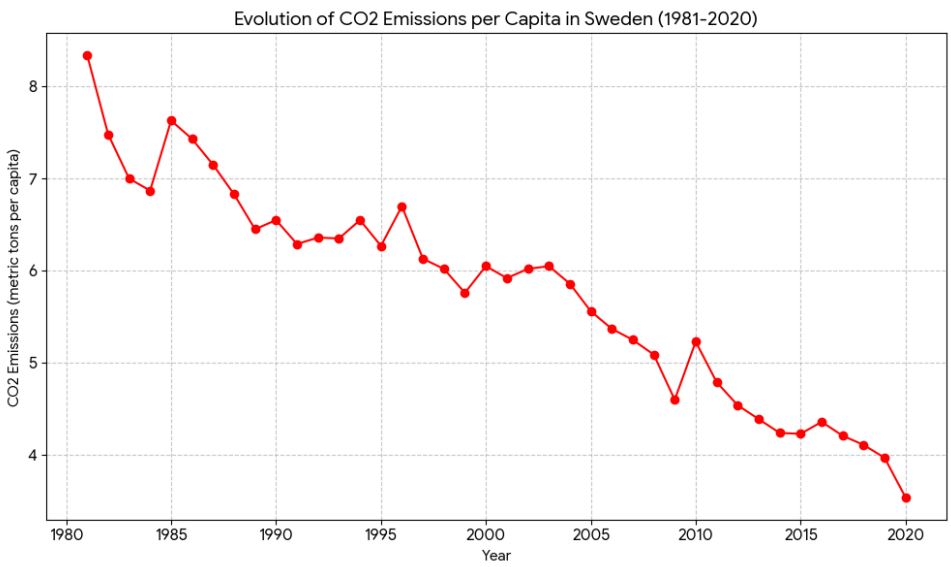

FIGURE 1 – Evolution of CO2 emissions per Capita (1981-2020)

This evolution suggests that carbon taxation played a strong incentive role by steering production and consumption behaviors toward less polluting energy choices.

6.2 Evolution of GDP per capita

Contrary to the concerns often associated with the implementation of a carbon tax, the Swedish economy did not experience a lasting slowdown in growth. GDP per capita shows an overall upward trend throughout the study period, with cyclical fluctuations linked to international economic crises. Between 1981 and 2020, GDP per capita more than tripled. This increase reflects a significant improvement in living standards and economic productivity.

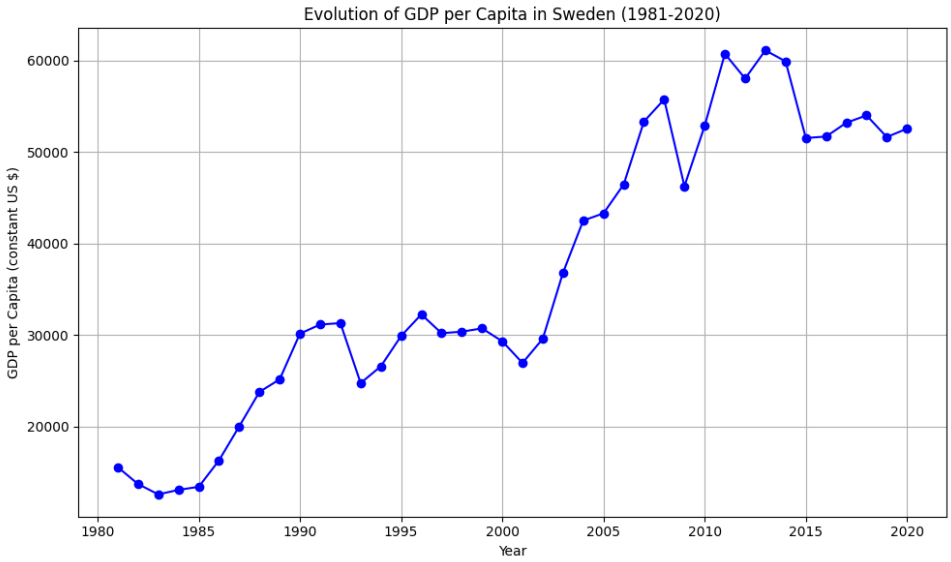

FIGURE 2 – Evolution of GDP per Capita (1981–2020)

The simultaneous occurrence of economic growth and declining emissions provides a first strong indication of a structural transformation of the development model.

6.3 Joint analysis of GDP and CO2 : Evidence of a structural break

The simultaneous observation of the two time series highlights a clear structural break in the early 1990s. Before 1991, economic growth was generally accompanied by high levels of emissions. Following the introduction of the carbon tax, this relationship weakened and eventually reversed.

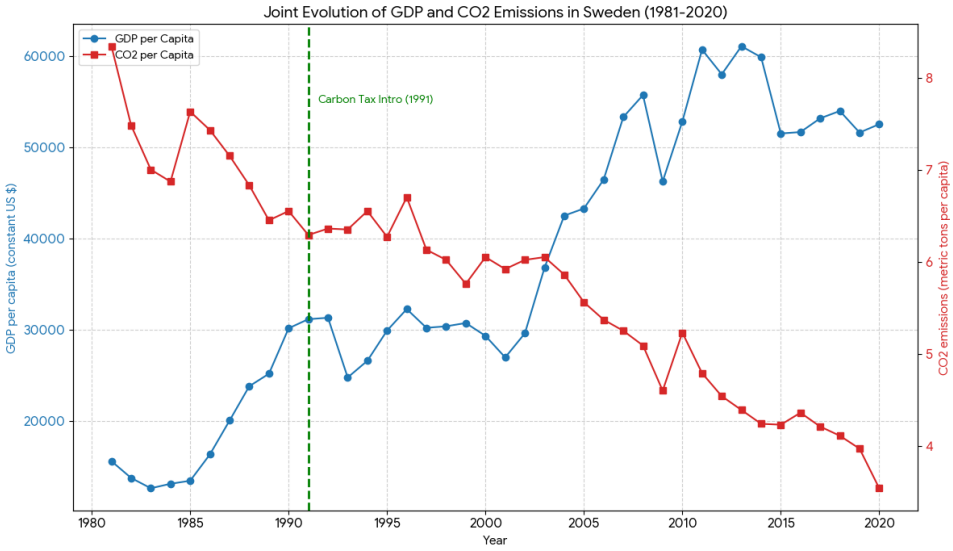

FIGURE 3 – Joint Evolution of GDP and CO2 Emissions with Structural Break (1991)

This break suggests that the carbon tax contributed to a profound transformation of the Swedish economy's energy and productive structure.

7 Statistical analysis and interpretation

7.1 Statistical comparison before and after the carbon tax

To more precisely assess the impact of the carbon tax, a comparison of average CO2 emissions before and after 1991 was conducted. The results show that average emissions per capita were significantly higher prior to the introduction of the tax than during the post-tax period. This decrease does not appear to be temporary but rather long-lasting, reinforcing the hypothesis of a structural effect of climate policy.

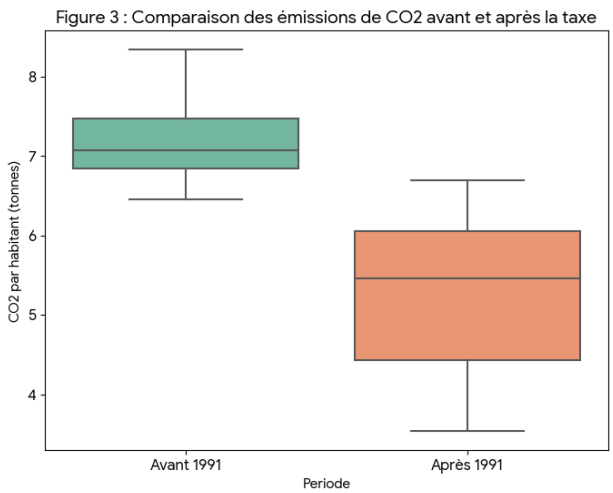

FIGURE 4 – Boxplot of CO2 Emissions Before and After 1991

The boxplot also highlights a reduction in the dispersion of emissions, suggesting a stabilization of the energy system.

7.2 Correlation analysis between GDP and CO2 emissions

The calculation of Pearson's correlation coefficient reveals a strong negative relationship between GDP per capita and CO2 emissions. This negative correlation indicates that, over the study period, increases in economic wealth were associated with decreases in emissions.

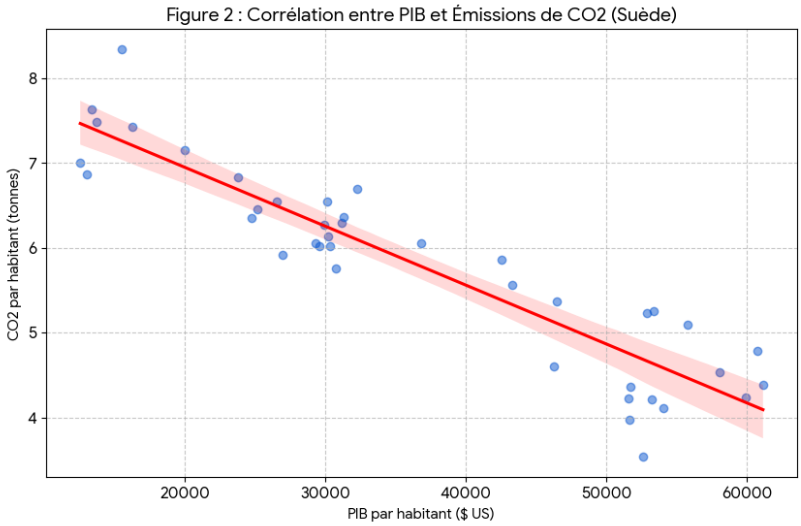

FIGURE 5 – Scatter Plot of GDP per capita and CO2 Emissions

This result provides empirical evidence of absolute decoupling, a phenomenon that remains rare among industrialized economies.

7.3 Economic interpretation of the results

The interpretation of these results suggests that the carbon tax altered economic incentives by making polluting activities more costly. Firms were thus encouraged to improve energy efficiency, while households gradually adjusted their consumption behavior.

Moreover, revenues generated by the carbon tax were used to finance investments in green infrastructure and clean technologies, thereby reinforcing the positive impact of climate policy.

8 Discussion of results

The findings confirm the predictions of Pigouvian theory, according to which taxing negative externalities helps correct market failures. They also align with the Environmental Kuznets Curve framework, showing that a high level of economic development can be compatible with declining emissions.

However, it should be emphasized that Sweden's success relies on a set of specific conditions, including a strong institutional framework and high social acceptance of environmental policies.

9 Limitations of the study

Despite its robust findings, this study has certain limitations. First, it relies on aggregated data, which does not allow for an analysis of sector-specific effects of the carbon tax. Second, other environmental policies implemented simultaneously may also have contributed to the observed reduction in emissions.

These limitations open avenues for further, more detailed research.

10 General conclusion and future perspectives

In conclusion, this mini-thesis demonstrates that a carbon tax can be an effective instrument in combating climate change without undermining economic growth. The Swedish experience thus provides a benchmark model for countries seeking to pursue an ambitious energy transition.

Future research perspectives may include cross-country comparative analyses or sectoral studies examining the differentiated impact of carbon taxation.

10.1 Policy recommendations

- Implement progressive and predictable carbon taxes in order to provide long-term visibility for economic agents ;
- Accompany carbon taxation with redistribution measures to mitigate social impacts and protect vulnerable households ;
- Reinvest tax revenues in the energy transition, particularly in renewable energy, energy efficiency, and green infrastructure.

References

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